SAM ADEYEMI

Backend, Web & AI Software Engineer · Python / FastAPI · PHP / Drupal · PostgreSQL · LLMs, RAG & Agentic AI

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PROFESSIONAL PROFILE

Backend and AI software engineer with 6+ years building secure, scalable backend services and shipping LLM-powered features to production, not just prototypes. Hands-on across the full backend: REST APIs and third-party integrations, PostgreSQL schema design and query optimisation on large event datasets, background jobs, real-time features, role-based access control and audit logging. On the AI side: OpenAI and Anthropic integration, structured outputs, RAG with embeddings and vector search, and multi-agent orchestration. Deploys self-hosted client software on Linux with Docker and Nginx, backed by automated testing and CI. On the web side: around 15 years building WordPress sites, and two council website projects in Drupal 11 and LocalGov Drupal, with custom modules, a WCAG 2.2 AA accessibility audit and automated accessibility checks. MSc Information Security & Software Engineering (Distinction), First Class BEng Software Engineering, peer-reviewed AI governance research, and a C#/ASP.NET and SQL Server foundation from university projects.

CORE TECHNICAL SKILLS

Languages	Python, SQL, PHP, TypeScript / JavaScript, C# (academic)
Backend & APIs	FastAPI, ASP.NET (academic), REST API design, third-party API integration, background job processing, real-time features, event-driven workflows
Data	PostgreSQL, Microsoft SQL Server, MySQL, pgvector; schema design, indexing and query optimisation on large event datasets
AI & LLM	OpenAI and Anthropic APIs: prompting, structured outputs, context handling; RAG, embeddings, vector and keyword search; evaluation sets, guardrails and cost controls; agentic and multi-agent orchestration
AI-assisted dev	Claude Code and GitHub Copilot in daily development
Security	Role-based access control, audit logging, API key management, secure-by-design APIs; AI governance (published research)
DevOps & quality	Docker, Nginx reverse proxy, Linux servers, self-hosted client deployments, unit and integration testing, CI pipelines, Git / GitHub
Web & CMS	Drupal 11 and LocalGov Drupal (custom modules, content entities, services, Form API, Twig theming), WordPress (around 15 years)
Accessibility	WCAG 2.2 AA audits and remediation, GOV.UK Design System patterns, automated checks with axe-core and Playwright
Frontend	React with TypeScript, Tailwind CSS, shadcn/ui

PROFESSIONAL EXPERIENCE

Lead Software & AI Systems Engineer | Apex Digital Africa **2019 – Present**

Web development and AI consultancy delivering backend, SaaS and AI products for clients and in-house platforms

- Design, build and maintain secure backend services, **REST APIs and third-party integrations** in Python and FastAPI, from requirements through testing, release and production support.
- Ship **LLM-powered features to production** using OpenAI and Claude APIs, with prompt design, structured outputs and context handling for domain-specific retrieval and workflow automation.
- Re-engineered project workflows around **event-driven and metadata-driven design** with schema-validated outputs, achieving **85% faster execution** and **73% fewer output errors**.
- Design PostgreSQL schemas, **indexes and queries for large event datasets**, and build **background jobs and real-time features** that keep dashboards and reports responsive.

- Implement **role-based access control, audit trails and reporting** for multi-user SaaS platforms, so every action is attributable.
- Deploy and run client software on **self-hosted Linux servers with Docker and Nginx**, backed by automated unit and integration tests and CI pipelines.
- Build React and TypeScript front ends with Tailwind CSS and shadcn/ui; explain technical trade-offs to non-technical clients without jargon.

SELECTED PROJECTS

Find My Service: Grounded AI Assistant *github.com/SamAde1203/find-my-service*

FastAPI service that answers residents' questions from a council's own pages using Claude, citing every source. Guardrails for emergencies, prompt injection and personal data; output checks on citations and links; no-AI fallback, daily cost cap and rate limiting. With Claude it passed all 53 evaluation questions, including 14 held out from tuning; 44 tests.

Python · FastAPI · Claude API · BM25 retrieval · pytest · mypy · Docker · GitHub Actions

Demo City Council on LocalGov Drupal *github.com/SamAde1203/localgov-council-microsite*

Council website on LocalGov Drupal 4 and Drupal 11. Built a custom module that adds a "Request an accessible format" link to every PDF and Word document, with a request form, reference numbers and a staff queue. Ran a WCAG 2.2 AA audit and fixed all 10 issues found, including error handling and focus management. 42 PHPUnit tests and axe-core checks run in CI.

PHP 8.3 · LocalGov Drupal · Drupal 11 · Twig · PHPUnit · Playwright / axe-core · GitHub Actions

Council Services on Drupal 11 *github.com/SamAde1203/council-services-drupal*

Custom Drupal 11 module and accessible theme: a postcode bin-day finder using the postcodes.io API, with caching, rate limiting and a fallback if the API is down. 26 PHPUnit tests and WCAG 2.2 AA checks in CI.

PHP 8.3 · Drupal 11 · Twig · PHPUnit · Playwright / axe-core · GitHub Actions · DDEV

PHHM Multi-Agent AI Platform *Lead Architect & Developer*

Architected an agentic platform where a central orchestrator routes work to five specialist agents (six agents in all) for communication, analytics, knowledge retrieval and user interaction, serving users across multiple countries.

Python · FastAPI · OpenAI API · REST · JSON / YAML · Linux

MR_CLAUDE_NOTEBOOK_LM: AI Engineering Operating System *In development*

Retrieval pipeline over technical documents (chunking, embeddings, vector storage and semantic search) behind a custom FastAPI backend, with answers validated against the retrieved evidence.

Python · FastAPI · Docker · PostgreSQL / pgvector

SeedTracker: AI-Powered Finance & Engagement SaaS *Founder & Lead Engineer*

Church finance platform live with 8 churches in Nigeria. Built the backend services, integrated payment and WhatsApp APIs, and designed RBAC, audit trails and AI-assisted financial reporting.

React / TypeScript · Node.js / NestJS · Python / FastAPI · PostgreSQL · Redis · Paystack

TrialForge AI & TDCT Validator *Lead Developer*

Built a rules-based validation engine running 20+ repeatable protocol-consistency checks, automating quality assurance and producing structured, auditable documentation.

Python · YAML

EDUCATION & RESEARCH

MSc Information Security & Software Engineering (Distinction) — **University of Greater Manchester**

BEng (Hons) Software Engineering (First Class Honours) — **University of Bolton**

- Built five data-driven web applications in **C# / ASP.NET** with **SQL Server** and MySQL back ends during the degree (library, learning, logistics, shopping and banking systems).
- Published peer-reviewed AI governance research in *AI and Ethics* (Springer Nature).
- Peer reviewer for the journal *Trials*.